

Artificial Intelligence Assisted Virtual Reality Warfighter Training System

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Abstract— Present military forces need to prepare for increasingly varied and complex situations and maintain the readiness of their warfighters for several different scenarios. There is a constant challenge of working with limited resources - people, time, and funding. Military leaders are seeking new training technologies that can meet their goals better, faster, and cheaper. Virtual Reality (VR) and Augmented Reality (AR) have emerged as extremely useful technologies for developing new training tools since they allow for easy creation and maintenance of multiple scenarios and environments. Using AR/VR for training lowers the travel and living costs associated with bringing the trainees to a central training facility, while still completely immersing the trainees in the environment. Another advantage of using AR/VR compared to traditional tools is that by using AR/VR, trainees are completely immersed in the environment, which makes it much more natural for them to consume the training material. At GE Research, we are exploring technologies to incorporate Artificial Intelligence (AI) methodologies in an immersive VR based training environment for warfighters. We have developed an AI assisted VR system for ground soldier training. In this VR training environment, the soldier (Blue Team/Blue Force) is teamed up with an AI assistant that will help them navigate an urban setting and successfully reach their goal in a stealth mode while escaping the gaze of enemy soldiers (Red Team/Red Force). We have used AI enabled virtual humans for the enemy soldiers. In addition, we have created an AI enabled, VR drone simulation to assist the dismounted soldier to move undetected through the environment.

Keywords—*Virtual Human, Blue Force, Red Force, Warfighter Training, Drone, Synthetic Environment.*

I. INTRODUCTION AND MOTIVATION

Virtual Reality (VR) and Augmented Reality (AR) technologies hold great promise and have seen an exponential rise in usage in different applications such as manufacturing, design, field services, healthcare, diagnostic imaging and training. Among them, training is one area that is especially suited to exploit the advantages of what VR/AR has to offer. VR/AR technologies are extremely useful as training tools since they allow for easy creation and maintenance of multiple scenarios and environments. This is especially true for large, complex systems such as aircraft structures, turbine engines, and warfighter training environments, where using real systems or creating scale model replicas with enough detail is prohibitively expensive. Using VR/AR for training lowers the travel and living costs associated with bringing the trainees to a central training facility. Another advantage of using VR/AR compared to traditional tools is that by using VR/AR trainees are virtually transported and immersed in the environment, which makes it

much more natural for them to consume the training material. On the industrial side, companies such as Boeing [1], Honeywell [2] and GE [3] have started to use VR/AR for training workers and customers. On the defense and military side, entities such as US Navy [4], US Air Force [5] and US Army [6] have also started using these technologies for training their warfighters and maintenance, repair and operations (MRO) personnel. The US Army has recently started building a comprehensive AR/VR based Synthetic Training Environment (STE) for training its warfighters for the different scenarios that they might face during a conflict [7]. The AR/VR based combat training application currently being developed by the Army will be used by multiple warfighters simultaneously across different training sites.

Although, there has been significant interest in recent years in developing VR/AR technologies for warfighter training, it is still very much an underdeveloped area. The integration of AI techniques with VR for training is also an area of active research. In this paper, we have tried to bridge this gap by developing an AI assisted, VR based, immersive synthetic environment to be used by dismounted soldiers in training for stealth-based situations. The focus of the current effort is for modeling and improving individual ground soldier operational effectiveness and survivability. For this paper, we have selected an urban environment wherein the user i.e. the Blue Force soldier must navigate from the starting location to the target location while avoiding the enemy i.e. the Red Force soldiers. The enemy Red Force soldiers are modeled as virtual humans and their behavior is enabled through AI. They act as a team and their main Course of Action (COA) is to prevent the blue force soldier to reach the end target. Within the VR environment, we have also developed a drone assistant application for supporting the Blue Force soldier. The drone assistant will help the user by providing aerial views and AI based suggestions of navigable locations, directions and best hiding spots.

For this project, we have used the Oculus Quest VR standalone headset and created the virtual environment on the Unreal Engine (UE) 4.24 platform. The Quest is a standalone device which does not need to be tethered to a computer. This provides improved convenience and freedom of movement. The Unreal Engine provides richly detailed graphics, high-speed processing performance, a built-in physics engine, and robust networking for shared environments. In the simulation, users are first transported to a tutorial level that provides instructions about how to use the controllers, how to utilize the drone AI assistant and the specifics of the training application, including the mission objective(s). Once the user indicates completion of

the tutorial level, they are transported to the main mission level. Here the user must complete the mission objective which is to make their way undetected from their starting location to the target location-indicated with a red X in the game. This level is completed through successfully arriving at the target location or failing to remain undetected by enemies in the environment. The details of the simulation environment development and implementation are described in the following sections.

II. SIMULATION IMPLEMENTATION

A. Virtual Environment

For this paper, we have created the virtual environment on the Unreal Engine platform and used the Oculus Quest VR standalone headset. We chose the Oculus Quest VR system for multiple reasons: i) does not require a high-end PC, ii) no wires, allowing for freedom of movement, iii) lower overall cost, and iv) 6 degrees of freedom (DOF). We chose Unreal engine because it has better graphics compared to its peers. We also plan to extend this demo and create more realistic environments and feel that UE4 will be more capable meeting our needs, especially for the next-gen VR devices. We were also interested in some of the features that UE4 provides, namely the blueprint and node-based visual scripting, complex particle simulations and advanced lighting. We leveraged several existing modules within the UE4 platform such as 1) Animation Starter Pack, 2) UE4 Mannequin, 3) Human Vocalizations, 4) Polygon City and Town Pack, 6) OculusVR Plugin and 7) Built in Pathing. We have used an urban setting as the training environment, however this training can be easily replicated for other environments such as semi-urban areas, deserts, forests, coasts, etc. The urban environment consists of nine blocks of a cityscape and we have utilized a low poly pack of buildings, props, vehicles, and environmental assets to create the city block areas. Within this environment, we have modeled several enemy soldiers as virtual humanoids who are constantly patrolling the environment. The player must avoid the enemy soldiers and is aided in their quest by an assistant drone, which they can initiate and fly overhead for aerial views. We attempted to make use of the built-in AI system, but it did not provide us with the tools to create the style of AI we desired. Therefore, we had to create our own version of AI, details are provided in the following sections.

B. AI Drone

The drone is a separate asset presented to the user in the tutorial and at the beginning of the mission. The user can choose whether to utilize the drone or not. Once in flight, the drone can cycle through 9 waypoints and provide the user with 10 points of view: a view from each waypoint and a view of the overall map. The drone's view of the environment can be seen through a heads up display (HUD) in the user's field of view (see Fig. 1) or through a handheld display (see Fig. 2). The drone view of the map will show a red arrow indicator for enemy position and direction, a blue arrow for the player's position and direction. In addition, we also show safe hiding spots where the user can go to avoid the gaze of the enemy soldiers. These safe spots or safe zones are indicated as yellow circles on the map and as yellow cones in the user's FOV, as seen in seen in Fig. 1 and Fig. 2. The safe spots are determined dynamically during the gameplay based on the positions of the user and the positions and gaze of the enemy AI. The safe spot algorithm is designed to either turn

on or turn off the safety cones in the environment (seen as circles by the drone above the city), depending upon the position of the player and the position and direction of the enemies. The safety cones are in predetermined positions throughout the urban environment.

The algorithm first checks to see if the player is close enough to the spot (6000 cm or about a city block) for it to be potentially activated (turned on) or too far away from the player's current position and is deactivated (turned off). If the player is within range of the specific safe spot, then we first look at the distance between the safe spot and the enemies. If there is no enemy within 7000 cm (just over the length of a city block) of the spot, then the spot is turned on, but if there is an enemy within 1000 cm, then the spot is turned off. If an enemy is within the 1000 to 7000 cm range, then a line trace is activated that originates at head height of the enemy and ends at the same height at the location of the safe spot. If the line trace is obstructed, which represents a game object obscuring the enemy's view of the spot, the cone is activated. If the line trace is unobstructed, then the enemy player's direction is determined, and a safety band is calculated. The safety band is 1000 cm on either side of the enemy and continuing in front of and behind the enemy. If the safe spot falls into this safety zone, then the spot is deactivated. If the spot is outside of this zone, then the final check begins. The algorithm compares the position of the player, enemy, and safe spot. A quadrant calculation is completed based on the position and direction of the enemy. The spot is deactivated if the safe spot's quadrant is in the direction the enemy is looking and the player and the safe spot are on opposite sides of the enemy. If the safe spot is in any of the other quadrants, then the safe spot is activated. At this point the algorithm is complete for this cycle. The algorithm for determining the safe spots are shown in the form of a flowchart in Fig. 3.



Fig. 1. HUD of Map



Fig. 2. Handheld Display of Map



Fig. 3. Flowchart for Creating Safe Zones

C. Enemy AI

We attempted to make use of the built-in AI system provided by UE, but the system did not provide us with the tools to create the style of AI we desired. The biggest deficiency was in terms of our desire to have the enemy sight to be fallible. To do this we needed to create our own player detection system which would not be simply black or white, seen or not seen, but rather have a level of ambiguity. In our implementation, there is a gray area where the enemy might have seen the player, but is not sure, so will call out to other nearby enemies and they will all go and check to see if the player is at the detected location.

The AI used by the enemy soldiers to detect the player utilizes a raycast algorithm. The enemy model continuously shoots five raycast vectors (line traces) and the enemy's response is determined by the percent of the raycasts that hits the user. TABLE 1 shows the distribution of raycasts and the corresponding enemy responses. Examples of these detections are shown in Fig. 4 and Fig. 5, where the cylindrical artifact is the collision mesh of the player. We have hidden the player mesh in these figures for better visualization of the concept. The rays originate from the eye location of the enemy and span directly vertically in front of the model for a 1.5 block length. The red color of the raycasts indicate that they have not intersected the user's mesh. If a ray intersects an object before the player, it turns green and is no longer counted. If the ray intersects the player without encountering any other object, it turns green and a red cube is placed at the point of intersection. These raycasts are only available in the developer mode and are not visible to the player. In the current implementation of the training, there are two types of enemy soldiers: a) Patrol and b) Lackey. Patrol enemies have a set path which they follow until specific stimulus is applied. Lackey enemies are similar, but instead randomly move about the environment rather than follow a given path. We created two different kinds of enemies for two reasons: 1) to create an environment where the player cannot learn the pattern over time and then go at the perfect moment, 2) to replicate the idea that the randomly moving enemy is investigating things that it believes it sees or hears and exploring places where the player could hide. The use of two types of enemies allows the objective to remain relatively safeguarded and prevents leverage of pattern recognition by the user. The objective cannot be reached with the same route or plan every time. The stimulus for the enemy soldiers is the detection of the user i.e. Blue Force soldier. While the player is 'unseen,' the enemy model has either a predetermined destination point (Patrol models) or a randomly selected point (Lackey models) in the navigable area. The navigable area is determined by the navmesh added into the playing environment. Fig. 6 shows this navmesh as the green overlay, and similar to the raycasts, the navmesh is only visible during the developer mode and not during the actual gameplay.

For the Lackey enemies, a random navigable point is determined within the navigable area and then the model proceeds to that point. For the Patrol enemies, predetermined points in the navigable area are provided and the model loops through these points. When an enemy model partially detects the player, it requests assistance from nearby enemies and changes its route to the location where it detected the player. Additionally, it will call for other enemy models in the area to join it; these models will then converge on the player's 'partial' detection location. The enemy behavior is determined according to TABLE 1. An example of this can be seen in Fig. 7. The enemy will continue moving towards the point where it first "sees" the user, irrespective of whether they player is still in that location or not. Once the enemy reaches that location and determines that the player has moved from that place, it will restart its regular movement based on whether it is a Lackey or a Patrol enemy model. A player can duck to avoid being seen by the enemy. When the player ducks, it lowers their visible profile and would lead to a 'partial' view and have enemies converging towards the user. While this will not keep a player from being spotted in the middle of the road, it will keep the player from being seen if they are able to duck behind a short obstacle, wall, object, etc.

D. Training Mission Gameplay

During the actual gameplay, the player will begin in the tutorial level. Here they will be taught how to use the drone AI assistant, the different view options, get familiarized with environment and understand the mission objective. The player can test out the button mapping and player movement using the controller joystick. The enemy models are inactive in this level which allows the player to familiarize themselves with the VR environment. When the player is done with the tutorial or if the player has already been through the tutorial level before, they can choose to move onto the mission level. Once in the mission level, the player will have access to the drone and the enemy models become active. As stated in previously, the player should strive to complete the mission objective which is to reach the target location undetected. There is an 8.5 second delay from the start of the mission level and when enemy models are spawned in the game. This delay is provided to allow time for the player to launch the drone without worry of interference from the enemy models. A successful mission is defined as the player being able to reach the target location, indicated with a red X, that the enemies are protecting. Upon doing so, they will be transported to a success level, shown in Fig. 8. A failure occurs when the player is fully seen by an enemy. Upon failure, the player is transported to a failure level.

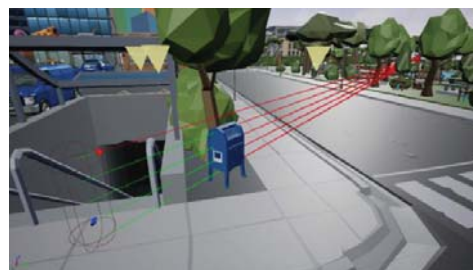


Fig. 4. Partially Seen User; 1 Ray Hit

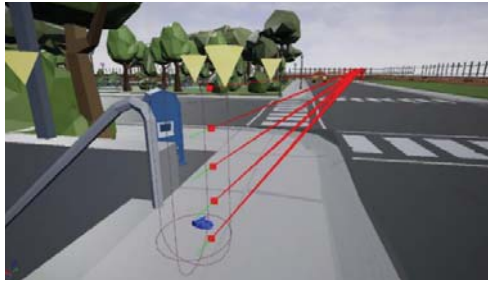


Fig. 5. Fully Seen User; All 5 Rays Hit

TABLE 1. RAYCAST HITS AND ENEMY RESPONSE

# Hits	Enemy Response
0	No response.
1-2	Partially sees the user. Calls for assistance by shouting "Contact" and goes to investigate.
3-5	Fully sees the user. Shouts "Attack... Fire" and the player fails the mission.



Fig. 6. Navigable Area/Navmesh

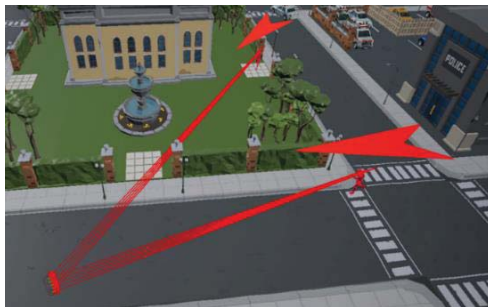


Fig. 7. Developer View of Enemy Convergence



Fig. 8. Snapshot of a Successful Mission

III. CONCLUSIONS AND FUTURE WORK

In this proposal, the team has demonstrated a VR platform for warfighter training. This training is only the first step in creating a full-fledged AI based immersive training environment. We are currently in the process of testing this training prototype both internally within GE Research and externally with our defense partners, including the DOD, Army, Navy and Air Force. In addition, these types of training systems can also be applied to industrial use cases such as factory walkthrough/flythrough, emergency evacuation, factory and warehouse planning, safety training, etc. There are several enhancements that we are planning for our system:

1. Increase the complexity of the enemy AI and behavior models to provide more variety and realism.
3. Implement a 'report card' feature and trainer AI to provide performance feedback to the user.
4. Include other types of environments such as forest, desert, etc. and expand the environment to a larger playing area.
5. Incorporate haptics and vibration feedback.
6. Create levels with increasing difficulties with variation in number of enemies, speed of enemies, etc.
7. Expand the current architecture to multiplayer system.
8. Add speech and audio functions to the system.

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